

EDUCATION

Harvard University , Cambridge, MA		2025 –
PhD, Electrical Engineering		
Advisor: Shriya S. Srinivasan, PhD		
Johns Hopkins University , Baltimore, MD	GPA: 4.00 / 4.00	2023 – 2025
MSE, Biomedical Engineering (Neuroengineering Focus)		
Thesis: Bridging Perception and Performance: Generating, Calibrating, & Evaluating Multimodal Sensory Stimuli		
Advisor: Jeremy D. Brown, PhD		
College of Engineering Pune , Pune, India	GPA: 8.24 / 10.00	2019 – 2023
BTech, Electronics and Telecommunication Engineering		

AWARDS AND HONORS

Harvard SEAS Prize Fellowship (\$281,796 • 6-yr tuition & 2-yr stipend) • Harvard University	2025 –
National Defense Science & Engineering Graduate Fellowship (\$277,484 • 3-yr tuition & stipend) • DOD	2025 –
GEM Associate Fellow • The National GEM Consortium	2025 –
Best Student Paper Finalist • International Conference on Rehabilitation Robotics (ICORR / RehabWeek)	2025
National Institute on Aging Start-Up Challenge Finalist (\$10,000) • NIH	2025
Best Paper Award • International Conference on Biomedical Engineering Science and Technology, India	2023
Mitacs Globalink Research Award (\$9,915) • Dalhousie University, Canada	2022
Best Project Award • IEEE National Project Competition, India	2021

RESEARCH EXPERIENCE

Johns Hopkins University • Haptics and Medical Robotics Lab ↗	Baltimore, MD
<i>Graduate Research Assistant</i> • Advisor: Jeremy D. Brown, PhD ↗	Jan 2024 – Jun 2025
Multimodal Visual-Haptic Neurorehabilitation for Post-Stroke Finger Dexterity	
Developed and evaluated a multimodal feedback framework for motor rehabilitation. Integrated cross-modal perceptual calibration to optimize sensory feedback delivery to individual thresholds. Implemented on ubiquitous devices to support smartphone-based rehabilitation outside the lab. <i>Publications:</i> P4 , C2 (Best Student Paper Finalist at ICORR 2025), C3 , A7	
Pneumatactors: Soft Multimodal Haptic Interfaces	
Engineered “pneumatactors” – a novel <u>pneumatic tactor</u> haptic interface designed for co-located stimulation of fast and slow adapting mechanoreceptors, enhancing cognitive salience. 3D-printed with flexible TPE to match mechanical impedance of human skin for intuitive interaction. <i>Publications:</i> P5 , A4 , A5	
Harvard University • Biohybrid Organs and Neuroprosthetics Lab ↗	Cambridge, MA
<i>Research Fellow</i> • Advisor: Shriya S. Srinivasan, PhD ↗	May 2024 – Aug 2024
Peripheral Nerve Stimulation to Modulate Cancer Tumor Metastasis	
Developed non-invasive peripheral nerve stimulation strategies to modulate tumor growth and metastasis in murine models, revealing frequency-dependent effects on tumor innervation. Designed and conducted pilot studies on melanoma-bearing mice, validating outcomes through immunohistochemical analysis.	
Surgical Approach for Treating Stress Urinary Incontinence	
Engineered miniaturized, RF-powered vagus nerve stimulators for freely moving rodents to study glioblastoma–nervous system interactions. Devices delivered monophasic pulses up to 1 kHz via custom PCB oscillators, enabling targeted parasympathetic modulation in vivo.	
IIT Bombay • Human Motor Neurophysiology and Neuromodulation Lab ↗	Mumbai, India
<i>Research Intern</i> • Advisor: Nivethida T, PhD ↗	Jan 2023 – Aug 2023

Quantifying Prospective Component of Sense of Agency using Machine Learning

Characterized pre-movement EEG signatures underlying the Sense of Agency using a modified Libet clock task with probabilistic auditory feedback. Applied spectral, time-series, and machine learning analyses to identify spatiotemporal patterns predictive of intentional binding. Demonstrated distinct neural correlates in pre-SMA and centroparietal regions, supporting a prospective model of agency. *Publications:* [A2](#), Bachelor's Dissertation

College of Engineering Pune • Departments of Electronics, Applied Sciences [↗](#)

Pune, India

Undergraduate Researcher • Advisor: Mahesh H. Shindikar, PhD [↗](#)

Aug 2022 – Aug 2023

Deep Learning-Based MRI Skull Tissue Segmentation

Developed and benchmarked vanilla, residual, and dense 2D U-Net architectures for skull stripping in neuroimaging, achieving 99.75% accuracy with a dense model optimized for feature reuse. Implemented data augmentation to improve generalizability across multi-scanner datasets. Demonstrated utility of open deep learning pipelines for replacing proprietary tools in MRI workflows. *Publications:* [C1](#) (Best Paper Award at ICBEST 2023), [A1](#)

Dalhousie University • Agricultural Mechanized Systems Group [↗](#)

Truro, Canada

Mitacs Globalink Research Intern • Advisor: Travis J. Esau, PhD [↗](#)

May 2022 – Aug 2022

Automating Harvesters to Reduce Operator Cognitive Load

Engineered a sensor-driven control system to automate bin switching and dynamic harvester head height adjustment, reducing operator load. Used RGB and ToF cameras for volumetric bin fill estimation ($\pm 10\%$ accuracy), and integrated crop height and GPS data via CAN bus to actuate harvesting arms in real time. System validated over two harvest days with positive field performance.

Queliz Lifetech • Early Stage Medtech Startup [↗](#)

Pune, India

Research Intern • Electronics Subsystem

Jun 2021 – Sep 2021

Point-of-Care Device for Finger Dexterity Rehabilitation in Acute Burn Patients

Designed an affordable, portable rehabilitation device to restore MCP, DIP, and PIP joint mobility in burn patients. Developed a precision-controlled actuation system with adaptive feedback to enable customized therapy in low-resource clinical settings.

JOURNAL PUBLICATIONS

- P5 **A. S. Pimpalkar** et al., "Pneumatactors: Soft 3D-printed interface for co-located transient and continuous tactile stimuli," in preparation, 2025.
- P4 **A. S. Pimpalkar**, A. M. West, D. Rai, J. Xu, and J. D. Brown, "Evaluation of multimodal feedback to augment motor control in precision grip," in preparation, 2025.
- P3 **A. S. Pimpalkar**, A. Slepian, and N. V. Thakor, "Vibrations at first contact encode object stiffness before grasp completion," *IEEE Sensors Letters* (in print), May 2025. Preprint: [arXiv](#) [↗](#)
- In Media: Featured in New Scientist Magazine. [Article](#) [↗](#)*
- P2 C. Harris, **A. S. Pimpalkar**, A. Aggarwal, J. Yang, X. Chen, S. Schmidgall, S. Rapuri, J. Greenstein, C. Overby-Taylor, and R. D. Stevens, "Risk prediction for non-cardiac surgery using the 12-lead electrocardiogram: An explainable deep learning approach," *British Journal of Anesthesiology* (in print), 2025. Preprint: [medRxiv](#) [↗](#)
- P1 **A. S. Pimpalkar** and D. Niture, "Towards contactless elevators with tinyML using person detection and keyword spotting," Jul 2024. Best Project Award. Preprint: [arXiv](#) [↗](#), [Code](#) [↗](#)

CONFERENCE PRESENTATIONS † presenter

FULL-LENGTH PAPERS

- C3 A. M. West, **A. S. Pimpalkar**, J. Xu, and J. D. Brown, "SenseMatch: Smartphone-based cross-modal matching for accessible perceptual assessments," *International Conference on Neural Engineering*, June 2025 (in review).
- C2 **A. S. Pimpalkar** †, A. M. West, J. Xu, and J. D. Brown, "Optimizing cross-modal matching for multimodal motor rehabilitation," *International Conference on Rehabilitation Robotics (ICORR / RehabWeek)*, May 2025. Best Student Paper Award Finalist. Podium presentation.
- C1 **A. S. Pimpalkar** †, R. K. Patole, K. D. Kamble, and M. H. Shindikar, "Performance evaluation of vanilla, residual, and dense 2D U-Net architectures for skull stripping of augmented 3D T1-weighted MRI head scans,"

ABSTRACTS AND SHORT ARTICLES

- A7 **A. S. Pimpalkar** †, D. Rai, J. U. Bartels, J. Xu, and J. D. Brown, “Visual-haptic feedback enhances finger individuation in a virtual precision grip neurotraining task,” *Biomedical Engineering Society (BMES)*, Oct 2024. Podium pres. [Abstract ↗](#)
- A6 H. Song † et al. [including **A. S. Pimpalkar**], “Towards visual function restoration through photoacoustic stimulation,” *Association for Research in Vision & Ophthalmology (ARVO)*, May 2024. Podium presentation. [Abstract ↗](#)
- A5 **A. S. Pimpalkar** †, P. Ameta †, A. Dalia, and J. D. Brown, “Pneumatactor arrays for high frequency vibrotactile feedback,” *IEEE Haptics Symposium*, Apr 2024. *Live demonstration and Work-in-Progress paper (2 presentations)*.
- A4 **A. S. Pimpalkar** †, P. Ameta †, A. Dalia †, and J. D. Brown, “Pneumatactor arrays for high frequency vibrotactile feedback,” *United States Senate AI Caucus: Robotics Demo Day*, Apr 2024. *Live demonstration*.
- A3 C. Harris †, **A. S. Pimpalkar**, A. Aggarwal, P. Yang, X. Chen, C. Overby-Taylor, J. Greenstein, and R. D. Stevens, “Surgical risk prediction using an explainable deep learning approach applied to pre-operative 12-lead electrocardiograms,” *Johns Hopkins Medicine & Engineering Research Retreat*, Feb 2024. *Poster presentation*. [Poster ↗](#)
- A2 **A. S. Pimpalkar** †, R. Patole, and N. Thirugnanasambandam, “Demonstrating the prospective component of sense of agency using machine learning,” *COEP ENT C BTEch Project Symposium*, May 2023. *Poster presentation*. [Poster ↗](#)
- A1 **A. S. Pimpalkar** †, R. Patole, K. Kamble, and M. Shindikar, “Evaluating U-Nets for skull stripping of augmented T1-weighted MRI scans,” *No Garland Neuroscience Conference*, Feb 2023. *Podium and poster presentation*. [Poster ↗](#)

TEACHING ENGAGEMENTS

Mechatronics, Johns Hopkins • Senior Level

Spring 2024

Teaching Assistant to Jeremy D. Brown, PhD

- Solved a 3-year challenge in real-time location polling of autonomous air hockey robots using CV-based ArUco tag detection and ZigBee mesh networking ([open-source implementation ↗](#)). Led labs on building autonomous robots. Worked in a mentorship role in Spring 2025.

Principles of the Design of Biomedical Instrumentation, Johns Hopkins • Graduate Level

Fall 2024

Teaching Assistant to Nitish V. Thakor, PhD

- Mentored labs on designing ECG, EMG, EEG recording circuits and projects including accessible gaming, wearables.

Haptic Interface Design for Human-Robot Interaction, Johns Hopkins • Graduate Level

Fall 2024

Class Researcher with Jeremy D. Brown, PhD

- Helped pilot an online learning platform to provide self-paced lectures incorporating live instructor interaction.

INVITED TALKS

Haptics: Neural Basis, Applications, Future Avenues. **Johns Hopkins University, Sep 2024. Neural & Rehabilitation Engineering Course**, invited by Nitish V. Thakor, PhD.

SKILLS AND CERTIFICATIONS

Medical Research	Human participant studies (non-invasive devices), rodent studies (invasive devices, mice/rats), human perception experiment design, IRB protocols
Electronics	Analog and digital circuit design, microcontroller programming (Arduino/Teensy/ESP/STM), high-speed and high-current PCB design (KiCAD/Altium), closed-loop control systems, DAQ design, FPGA, RF design, precision soldering (THT/SMT), mechatronic integration, communication protocols

Programming & CS	Python, MATLAB, C/C++, data analysis, VR rendering, machine and deep learning, TinyML, TensorFlow (+lite/micro), computer vision, LaTeX
Fabrication & Design	CAD (Autodesk, Shapr3D), FDM 3D printing (rigid/flexible), SLA resin printing, silicone molding
Certifications ↗ (15 total courses)	Harvard Online • TinyML Certificate (3 courses), Python for Research, CS50 Web Development MIT Online • Aeronautics and Human Spaceflight Johns Hopkins Online • Human Subject Research, Neuroimaging, Neurohacking DeepLearning.AI • Deep Learning Specialization (5 courses) AICTE • Computer Science & Biology (faculty development program completed with permission)

ACADEMIC SERVICE

IEEE Haptics Symposium • Student Volunteer	2024
Canadian Society of Bioengineering Annual Meeting • Student Volunteer	2022

LEADERSHIP AND TEAMWORK

Johns Hopkins India Institute • Student Leader, Operations and Logistics	2024 – 2025
• Managed operations, logistics, sponsorships for university's first Hopkins India Conference, Washington DC.	
Johns Hopkins Biomedical Engineering • Master's Program Representative	2024 – 2025
COEP Impressions Cultural Festival • Head of Events (2021), Volunteer	2019 – 2021
• Led the planning and execution of 28 annual events with 10k attendees, managing a ₹2m budget and coordinated with an overall team of 126 members to execute concerts, competitions, and international celebrity engagements.	
• Built and led a team of 17, working with 2 co-heads to build a platform from the ground up, fostering opportunities for budding artists across India and uniting them under a shared vision. Placed emphasis on unconventional art forms.	
Rowing Team • COEP, Pune City • Competitive 1X, 2X, 4X Sculler	2020 – 2022
• Won multiple race events and an institutional award commending motivation and skill. Represented city at regional races.	
COEP Mental Health & Wellness Center • Student Volunteer	2020 – 2022
COEP Data Science & AI Club • Research Lead (2020-21), Member	2020 – 2022